

For the sender, type:

```
nc -lp 1234 < testfile.txt
```

where 1234 is the port number and *testfile.txt* is the file to be transferred.

For the receiver, type:

```
nc 192.168.56.103 1234 > received.txt
```

2. If you have a lot of files, you can zip them into a single file and then send the zipped file over the network using Netcat. To transfer all the HTML files in a folder, use the following commands.

For the sender, type:

```
tar zc *.html | nc -lp 1234
```

For the receiver, type:

```
nc 192.168.56.103 1234 | tar zx
```

 **Note:** Sender and receiver commands can be used interchangeably. If you have to copy files from one computer to other computers on a network, you could use the above method in a loop.

For the sender, type:

```
while true; tar zc * | nc -lp 1234 ; done
```

Netcat overwrites a file with the same name as mentioned. Make sure that you save the receiving file with a temporary name and then try renaming it to avoid unexpected overwrites.

**Banner grabbing:** Banner grabbing is a technique to get information about a service running in an open port.

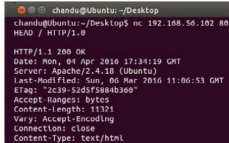
For example, if you know a service is running on a specific port on a specific machine, you could try giving commands (which may not be correct). The machine at the other end will send you a response stating that your request is either good or bad. In this response, you can actually find out the name of the service running on the port. This functionality of Netcat is more useful to network administrators who are monitoring the entire network for malicious activities.

```
nc <IP ADDRESS> <PORT>
```

You can give any command to get the response.

If you are trying to get the service name running on port 80 (http service), try giving the command *HEAD HTTP/1.0* and you will get the response with the name of the service utilising port 80 [Figure 3].

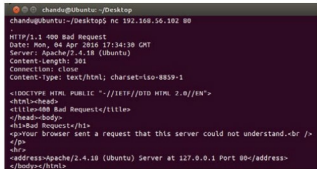
You can also give a random command, and you will get a 'BAD REQUEST' response with the banner of the service.



```
chandu@Ubuntu: ~/Desktop
chandu@Ubuntu:~/Desktop$ nc 192.168.56.102 80
HEAD / HTTP/1.0

HTTP/1.1 200 OK
Date: Mon, 04 Apr 2016 17:34:19 GMT
Server: Apache/2.4.18 (Ubuntu)
Last-Modified: Sun, 06 Mar 2016 11:06:53 GMT
ETag: "2c39-52d5f84b360"
Accept-Ranges: bytes
Content-Length: 11321
Vary: Accept-Encoding
Connection: close
Content-Type: text/html
```

Figure 3: Http response



```
chandu@Ubuntu:~/Desktop
chandu@Ubuntu:~/Desktop$ nc 192.168.56.102 80
HTTP/1.1 400 Bad Request
Date: Mon, 04 Apr 2016 17:34:30 GMT
Server: Apache/2.4.18 (Ubuntu)
Content-Length: 301
Connection: close
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>400 Bad Request</title>
</head><body>
<h1>Bad Request</h1>
<p>Your browser sent a request that this server could not understand.<br />
</p>
<hr/>
<address>Apache/2.4.18 (Ubuntu) Server at 127.0.0.1 Port 80</address>
</body></html>
```

Figure 4: 'Bad request' response

From Figures 3 and 4 one can find that the service running on port 80 is Apache/2.4.18 and the base system OS is Ubuntu.

**Getting remote shell:** Netcat has made it so easy for network admins to access the terminal of various systems, remotely. To access the remote terminal, the listener (system whose terminal should be remotely accessed) should use the following command:

```
nc -lp <port> -e /bin/bash
```

 **Note:** This functionality is not present in OpenBSD. The above command works only with the Netcat-traditional version.

Then, to access the remote shell, just connect to the listener using the following command:

```
nc <IP ADDRESS> <PORT>
```

Netcat can be used for a lot of other purposes such as transferring images or data files. For Windows users the above commands get slightly changed. You can even try to code your own program which can automate your daily network management tasks. 

**By: B.N. Chandrapal**

The author is a security enthusiast and has a keen interest in AppSec, robotics and open source software. He is currently studying for his B.Tech at Christ University, Bengaluru. He can be contacted at [chandrapal@linux.com](mailto:chandrapal@linux.com).